MPG No. 4.1.9 m [Type I, monoclinic] [G tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & G_y & 0 \end{bmatrix}$$

$$G_y = \mathbb{G}_y^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ G_{xy} & 0 & G_{yz} \\ 0 & G_{yz} & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{G}_{xy}^{(1)}$$
$$G_{yz} = \mathbb{G}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ -G_{xy} & 0 & G_{yz} \\ 0 & -G_{yz} & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{Q}_z^{(1)}$$
$$G_{yz} = \mathbb{Q}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & G_{yyy} & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & G_{zzy} \\ 0 & G_{yzx} & 0 \\ G_{xxy} & 0 & G_{yzx} \end{bmatrix}$$

$$G_{xxy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\beta(1)}$$
$$G_{yyy} = 3\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)}$$
$$G_{zzy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\beta(1)}$$
$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzy} & 0 \\ G_{yzx} & 0 & -\frac{G_{zzy}}{2} \\ 0 & G_{zxy} & 0 \\ -\frac{G_{xxy}}{2} & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{xxy} = 2\mathbb{G}_y^{(2)} - 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{zzy} = 2\mathbb{G}_y^{(2)} + 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{yzx} = 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$\frac{G_{zzy}}{2} = \mathbb{G}_y^{(2)} + \mathbb{Q}_{xz}^{(1)}$$
$$G_{zxy} = -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$\frac{G_{xxy}}{2} = \mathbb{G}_y^{(2)} - \mathbb{Q}_{xz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & G_{yzz} \\ 0 & G_{zxy} & 0 \\ G_{xyx} & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$
$$G_{yzz} = -\mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{zxy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$
$$G_{xyx} = \mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & G_{xxyz} & 0 & G_{xxxy} \\ 0 & 0 & 0 & G_{yyyz} & 0 & G_{yyxy} \\ 0 & 0 & 0 & G_{zzyz} & 0 & G_{zzxy} \\ G_{xxyz} & G_{yyyz} & G_{zzyz} & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{yzzx} & 0 & G_{zxxy} \\ G_{xxxy} & G_{yyxy} & G_{zzxy} & 0 & G_{zxxy} & 0 \end{bmatrix}$$

$$G_{xxyz} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)}$$
$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)}$$
$$G_{yyyz} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$
$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)}$$
$$G_{zzyz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$
$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)}$$
$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)}$$
$$G_{zxxy} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & G_{xxyz} & 0 & G_{xxxy} \\ 0 & 0 & 0 & G_{yyyz} & 0 & G_{yyxy} \\ 0 & 0 & 0 & G_{zzyz} & 0 & G_{zzxy} \\ -G_{xxyz} & -G_{yyyz} & -G_{zzyz} & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & -G_{yzzx} & 0 & G_{zxxy} \\ -G_{xxxy} & -G_{yyxy} & -G_{zzxy} & 0 & -G_{zxxy} & 0 \end{bmatrix}$$

$$G_{xxyz} = \mathbb{G}_{yz}^{(3)} - 4\mathbb{Q}_x^{\beta(1)}$$
$$G_{xxxy} = \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_z^{(1)} - 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$
$$G_{yyyz} = \mathbb{G}_{yz}^{(3)} + 2\mathbb{Q}_x^{(1)} - 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)}$$
$$G_{yyxy} = \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$
$$G_{zzyz} = \mathbb{G}_{yz}^{(3)} - 2\mathbb{Q}_x^{(1)} + 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)}$$
$$G_{zzxy} = \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_z^{\beta(1)}$$
$$G_{yzzx} = -\mathbb{Q}_z^{(1)} - 4\mathbb{Q}_z^{\alpha(1)}$$
$$G_{zxxy} = -\mathbb{Q}_x^{(1)} - 4\mathbb{Q}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ G_{yzzx} & 0 & G_{zxxy} \\ 0 & G_{zxxy} & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}^{(4)}_{xy}$$
$$G_{zxxy} = 2\mathbb{G}^{(4)}_{yz}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ -G_{yzzx} & 0 & G_{zxxy} \\ 0 & -G_{zxxy} & 0 \end{bmatrix}$$

$$G_{yzzx} = -\mathbb{Q}^{(2)}_{z}$$
$$G_{zxxy} = -\mathbb{Q}^{(2)}_{x}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} G_{xxyz} & 0 & G_{xxxy} \\ G_{yyyz} & 0 & G_{yyxy} \\ G_{zzyz} & 0 & G_{zzxy} \\ 0 & G_{yzzx} & 0 \\ G_{zxyz} & 0 & G_{zxxy} \\ 0 & G_{xyzx} & 0 \end{bmatrix}$$

$$G_{xxyz} = \mathbb{Q}^{(3)}_{x} + 2\mathbb{Q}^{(4)}_{x} + 2\mathbb{Q}^{\alpha(2)}_{x}$$
$$G_{xxxy} = -2\mathbb{G}^{(5)}_{xy} + \mathbb{Q}^{(3)}_{z} - \mathbb{Q}^{\alpha(2)}_{z} + \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{yyyz} = -2\mathbb{G}^{(5)}_{yz} + \mathbb{Q}^{(3)}_{x} - \mathbb{Q}^{\alpha(2)}_{x} + \mathbb{Q}^{\beta(2)}_{x}$$
$$G_{yyxy} = 2\mathbb{G}^{(5)}_{xy} + \mathbb{Q}^{(3)}_{z} - \mathbb{Q}^{\alpha(2)}_{z} - \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{zzyz} = 2\mathbb{G}^{(5)}_{yz} + \mathbb{Q}^{(3)}_{x} - \mathbb{Q}^{\alpha(2)}_{x} - \mathbb{Q}^{\beta(2)}_{x}$$
$$G_{zzxy} = \mathbb{Q}^{(3)}_{z} + 2\mathbb{Q}^{(4)}_{z} + 2\mathbb{Q}^{\alpha(2)}_{z}$$
$$G_{yzzx} = -\mathbb{G}^{(5)}_{xy} + \mathbb{Q}^{(4)}_{z} - \mathbb{Q}^{\alpha(2)}_{z} - \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{zxyz} = \mathbb{G}^{(5)}_{xy} + \mathbb{Q}^{(4)}_{z} - \mathbb{Q}^{\alpha(2)}_{z} + \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{zxxy} = -\mathbb{G}^{(5)}_{yz} + \mathbb{Q}^{(4)}_{x} - \mathbb{Q}^{\alpha(2)}_{x} - \mathbb{Q}^{\beta(2)}_{x}$$
$$G_{xyzx} = \mathbb{G}^{(5)}_{yz} + \mathbb{Q}^{(4)}_{x} - \mathbb{Q}^{\alpha(2)}_{x} + \mathbb{Q}^{\beta(2)}_{x}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} G_{yzxx} & G_{yzyy} & G_{yzzz} & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & G_{zxxy} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & G_{xyzx} & 0 \end{bmatrix}$$

$$G_{yzxx} = \mathbb{Q}^{(5)}_{x} + 2\mathbb{Q}^{(6)}_{x} + 2\mathbb{Q}^{\alpha(3)}_{x}$$
$$G_{yzyy} = -2\mathbb{G}^{(6)}_{yz} + \mathbb{Q}^{(5)}_{x} - \mathbb{Q}^{\alpha(3)}_{x} + \mathbb{Q}^{\beta(3)}_{x}$$
$$G_{yzzz} = 2\mathbb{G}^{(6)}_{yz} + \mathbb{Q}^{(5)}_{x} - \mathbb{Q}^{\alpha(3)}_{x} - \mathbb{Q}^{\beta(3)}_{x}$$
$$G_{yzzx} = \mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{(6)}_{z} - \mathbb{Q}^{\alpha(3)}_{z} + \mathbb{Q}^{\beta(3)}_{z}$$
$$G_{zxyz} = -\mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{(6)}_{z} - \mathbb{Q}^{\alpha(3)}_{z} - \mathbb{Q}^{\beta(3)}_{z}$$
$$G_{zxxy} = \mathbb{G}^{(6)}_{yz} + \mathbb{Q}^{(6)}_{x} - \mathbb{Q}^{\alpha(3)}_{x} + \mathbb{Q}^{\beta(3)}_{x}$$
$$G_{xyxx} = -2\mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{(5)}_{z} - \mathbb{Q}^{\alpha(3)}_{z} + \mathbb{Q}^{\beta(3)}_{z}$$
$$G_{xyyy} = 2\mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{(5)}_{z} - \mathbb{Q}^{\alpha(3)}_{z} - \mathbb{Q}^{\beta(3)}_{z}$$
$$G_{xyzz} = \mathbb{Q}^{(5)}_{z} + 2\mathbb{Q}^{(6)}_{z} + 2\mathbb{Q}^{\alpha(3)}_{z}$$
$$G_{xyzx} = -\mathbb{G}^{(6)}_{yz} + \mathbb{Q}^{(6)}_{x} - \mathbb{Q}^{\alpha(3)}_{x} - \mathbb{Q}^{\beta(3)}_{x}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & G_{xxyz} & 0 & G_{xxxy} & G_{xxzy} & 0 & G_{xxyx} \\ 0 & 0 & 0 & G_{yyyz} & 0 & G_{yyxy} & G_{yyzy} & 0 & G_{yyyx} \\ 0 & 0 & 0 & G_{zzyz} & 0 & G_{zzxy} & G_{zzzy} & 0 & G_{zzyx} \\ G_{yzxx} & G_{yzyy} & G_{yzzz} & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & G_{zxxy} & G_{zxzy} & 0 & G_{zxyx} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & G_{xyzx} & 0 & 0 & G_{xyxz} & 0 \end{bmatrix}$$

$$G_{xxyz} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)} + \mathbb{Q}_{x}^{(3)} + 2\mathbb{Q}_{x}^{(4)} + 2\mathbb{Q}_{x}^{\alpha(2)} - 4\mathbb{Q}_{x}^{\beta(1)}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} + 2\mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(3)} - 2\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} + \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{xxzy} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)} - \mathbb{Q}_{x}^{(3)} - 2\mathbb{Q}_{x}^{(4)} - 2\mathbb{Q}_{x}^{\alpha(2)} - 4\mathbb{Q}_{x}^{\beta(1)}$$

$$G_{xxyx} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} + 2\mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(3)} - 2\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} - \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{yyyz} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} + 2\mathbb{Q}_{x}^{(1)} + \mathbb{Q}_{x}^{(3)} - 2\mathbb{Q}_{x}^{\alpha(1)} - \mathbb{Q}_{x}^{\alpha(2)} + 2\mathbb{Q}_{x}^{\beta(1)} + \mathbb{Q}_{x}^{\beta(2)}$$

$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - 2\mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(3)} + 2\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} - \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{yyzy} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)} + 2\mathbb{Q}_{x}^{(1)} - \mathbb{Q}_{x}^{(3)} - 2\mathbb{Q}_{x}^{\alpha(1)} + \mathbb{Q}_{x}^{\alpha(2)} + 2\mathbb{Q}_{x}^{\beta(1)} - \mathbb{Q}_{x}^{\beta(2)}$$

$$G_{yyyx} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - 2\mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(3)} + 2\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} + 2\mathbb{Q}_{z}^{\beta(1)} + \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{zzyz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)} - 2\mathbb{Q}_{x}^{(1)} + \mathbb{Q}_{x}^{(3)} + 2\mathbb{Q}_{x}^{\alpha(1)} - \mathbb{Q}_{x}^{\alpha(2)} + 2\mathbb{Q}_{x}^{\beta(1)} - \mathbb{Q}_{x}^{\beta(2)}$$

$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + \mathbb{Q}_{z}^{(3)} + 2\mathbb{Q}_{z}^{(4)} + 2\mathbb{Q}_{z}^{\alpha(2)} - 4\mathbb{Q}_{z}^{\beta(1)}$$

$$G_{zzzy} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} - 2\mathbb{Q}_{x}^{(1)} - \mathbb{Q}_{x}^{(3)} + 2\mathbb{Q}_{x}^{\alpha(1)} + \mathbb{Q}_{x}^{\alpha(2)} + 2\mathbb{Q}_{x}^{\beta(1)} + \mathbb{Q}_{x}^{\beta(2)}$$

$$G_{zzyx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} - \mathbb{Q}_{z}^{(3)} - 2\mathbb{Q}_{z}^{(4)} - 2\mathbb{Q}_{z}^{\alpha(2)} - 4\mathbb{Q}_{z}^{\beta(1)}$$

$$G_{yzxx} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} - \mathbb{G}_{yz}^{(3)} + 4\mathbb{Q}_{x}^{\beta(1)}$$

$$G_{yzyy} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)} - 2\mathbb{Q}_{x}^{(1)} + 2\mathbb{Q}_{x}^{\alpha(1)} - 2\mathbb{Q}_{x}^{\beta(1)}$$

$$G_{yzzz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)} + 2\mathbb{Q}_{x}^{(1)} - 2\mathbb{Q}_{x}^{\alpha(1)} - 2\mathbb{Q}_{x}^{\beta(1)}$$

$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} - \mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(4)} - 4\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} - \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{yzxz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} - \mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(4)} - 4\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} + \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{zxyz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(1)} + \mathbb{Q}_{z}^{(4)} + 4\mathbb{Q}_{z}^{\alpha(1)} - \mathbb{Q}_{z}^{\alpha(2)} + \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{zxxy} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(5)} - \mathbb{Q}_{x}^{(1)} + \mathbb{Q}_{x}^{(4)} - 4\mathbb{Q}_{x}^{\alpha(1)} - \mathbb{Q}_{x}^{\alpha(2)} - \mathbb{Q}_{x}^{\beta(2)}$$

$$G_{zxzy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_{z}^{(1)} - \mathbb{Q}_{z}^{(4)} + 4\mathbb{Q}_{z}^{\alpha(1)} + \mathbb{Q}_{z}^{\alpha(2)} - \mathbb{Q}_{z}^{\beta(2)}$$

$$G_{zxyx} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(5)} - \mathbb{Q}_{x}^{(1)} - \mathbb{Q}_{x}^{(4)} - 4\mathbb{Q}_{x}^{\alpha(1)} + \mathbb{Q}_{x}^{\alpha(2)} + \mathbb{Q}_{x}^{\beta(2)}$$

$$G_{xyxx} = \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_{z}^{(1)} + 2\mathbb{Q}_{z}^{\alpha(1)} - 2\mathbb{Q}_{z}^{\beta(1)}$$

$$G_{xyyy} = -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_{z}^{(1)} - 2\mathbb{Q}_{z}^{\alpha(1)} - 2\mathbb{Q}_{z}^{\beta(1)}$$

$$G_{xyzz} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} - \mathbb{G}_{xy}^{(3)} + 4\mathbb{Q}_{z}^{\beta(1)}$$

$$G_{xyzx} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(5)} + \mathbb{Q}_{x}^{(1)} + \mathbb{Q}_{x}^{(4)} + 4\mathbb{Q}_{x}^{\alpha(1)} - \mathbb{Q}_{x}^{\alpha(2)} + \mathbb{Q}_{x}^{\beta(2)}$$

$$G_{xyxz} = 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(5)} + \mathbb{Q}_{x}^{(1)} - \mathbb{Q}_{x}^{(4)} + 4\mathbb{Q}_{x}^{\alpha(1)} + \mathbb{Q}_{x}^{\alpha(2)} - \mathbb{Q}_{x}^{\beta(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} G_{yzxx} & G_{yzyy} & G_{yzzz} & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & G_{zxxy} & G_{zxzy} & 0 & G_{zxyx} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & G_{xyzx} & 0 & 0 & G_{xyxz} & 0 \end{bmatrix}$$

$$
\begin{aligned}
G_{yzxx} &= \mathbb{Q}_x^{(5)} + 2\mathbb{Q}_x^{(6)} + 2\mathbb{Q}_x^{\alpha(3)} \\
G_{yzyy} &= -2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{yzzz} &= 2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{yzzx} &= 2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzxz} &= -2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{zxyz} &= 2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxxy} &= 2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{zxzy} &= -2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxyx} &= -2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{xyxx} &= -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{xyyy} &= 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyzz} &= \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)} \\
G_{xyzx} &= 2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{xyxz} &= -2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & G_{xxxy} & 0 \\
G_{yyyx} & 0 & G_{yyyz} \\
0 & G_{zzzy} & 0 \\
0 & G_{yyzy} & 0 \\
0 & G_{zzxy} & 0 \\
G_{xxyx} & 0 & G_{xxyz} \\
G_{yzzx} & 0 & G_{yzzz} \\
0 & G_{zzxy} & 0 \\
0 & G_{xyyy} & 0 \\
-\frac{G_{yyyz}}{2} + \frac{G_{yyzy}}{2} + G_{zxxy} & 0 & \frac{G_{xxxy}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} - G_{yyyx} + G_{zzxy}
\end{bmatrix}
$$

$$
\begin{aligned}
G_{xxxy} &= \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} \\
G_{yyyx} &= -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} \\
G_{yyyz} &= \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} \\
G_{zzzy} &= -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} \\
G_{yyzy} &= \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)} \\
G_{zzxy} &= 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} \\
G_{xxyx} &= \mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} \\
G_{xxyz} &= 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)} \\
G_{yzzx} &= 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} \\
G_{yzzz} &= -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)} \\
G_{zxxy} &= 2\mathbb{G}_{4x}^{\beta(1)} + \mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(5)} \\
G_{xyyy} &= -\mathbb{G}_{4z}^{\alpha(1)} - \mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)}
\end{aligned}
$$